

035 Env Chemgen Vanacloig Cgt

2026.09.29 - Vanacloig alone, compound-cold: design and the first finished arm

The first modeling experiment of the inhibitor-tolerance line. The question is whether the cell graph transformer predicts Vanacloig 2022 for a compound it has not seen once the compound reaches the readout. Plan: `notes-tex/031-unified-representation` (PR #436). Store check: `notes-tex/033-post-query-analysis` (PR #455). This branch is PR #473.

Inputs, by path. The cell table of experiment 033 (`$DATA_ROOT/experiments/033-env-chemgen-pooled/cell_table/cell_table.parquet`, slurm 2988) and the FCFP4 table of experiment 031 (`results/embeddings/fcfp4_count.npz`). 143,218 cells, 3,598 queried genes, 41 compounds.

Strain. Every strain is the queried deletion plus the three host deletions (PDR1, PDR3, SNQ2), so the perturbation set has four genes and three are constant. The library is not BY4741: Piotrowski 2017 builds the host deletions in the SGA query strain Y7092 (Y13206) and crosses it to the MATa deletion array, so each strain is a haploid segregant of two S288C-derived parents. The selection Piotrowski describes (canavanine and thialysine) implies the query strain's `can1` and `lyp1` deletions are present as well, which the loader does not record, and no SGA-derived loader does. The background is constant across the library, so none of this changes a Vanacloig-only result.

Split and score. Five grouped folds over compounds (`[[experiments.035-env-chemgen-vanacloig-cgt.scripts.vanacloig_data]]`); every compound is tested once; four validation compounds per fold pick the epoch. Per held-out compound, Spearman over its genes, raw and centered, beside the compound's ceiling. Each side of the centered score is centered by its OWN mean over the training compounds. Subtracting the measured gene mean from the prediction rewarded an untrained model: 0.356 centered and -0.001 raw at 60 steps (slurm 3009).

Same-fold references (`[[experiments.035-env-chemgen-vanacloig-cgt.scripts.baselines_same_folds]]`), median Spearman over the 41 test compounds, 28 or 29 training compounds per fold:

model	raw	centered
gene mean	0.209	undefined
ridge on FCFP4	0.286	0.316

These agree with the 031 leave-one-compound-out values (0.212, 0.311).

Arms (`[[experiments.035-env-chemgen-vanacloig-cgt.scripts.train_vanacloig_cgt]]`): the encoder is used unchanged and the readout is built in the experiment, `[h_CLS ; z_S ; z_E]` with `z_S` the sum of the perturbed embeddings over the four genes and `z_E` a projection of `log1p` of the fingerprint. Model half from the 025 replication config; batch 256, 12 epochs, AdamW with a one-cycle schedule to 5e-4. Batch 512 with the graph prior ran out of memory at 44 GB (slurm 3010).

arm	launch	slurm	state on 2026.09.29 16:40
<code>embedding_env</code> , no encoder	CPU	3020	finished, 5 folds
<code>cgt_env</code> , graph prior 1	GPU	3016	partial, fold 0 at epoch 2 of 12
<code>cgt_env</code> , graph prior 0	GPU	3017	partial, fold 0 at epoch 3 of 12
<code>cgt_genes</code> , floor, folds 0 and 1	GPU	3019	queued

Finished: the embedding table is below ridge. `embedding_env` over all 41 test compounds at the validation-selected epoch (`[[experiments.035-env-chemgen-vanacloig-cgt.scripts.summarize_round]]`, `results/round_summary.csv`): median Spearman 0.134 centered against ridge 0.316 on the same

compounds, above ridge on 14 of 41; 0.190 raw against ridge 0.286 and gene mean 0.209. Per fold the centered median runs from 0.005 to 0.318. Validation loss is unstable: the last epoch’s validation MSE is 9.12 on fold 3 and 1.12 on fold 1 against 0.30 to 0.54 on the others.

Observed, not explained. Sodium glyoxylate has a reliability index at or below zero, so its ceiling is recorded as zero, and both the embedding arm (0.402) and ridge (0.316) score it well above zero on the centered target.

2026.09.29 - Without the graph prior the encoder collapses every gene token

Measured by `[[experiments.035-env-chemgen-vanacloig-cgt.scripts.diagnose_token_collapse]]` (slurm 3024, `results/token_collapse.csv`) on 512 sampled genes and 512 sampled strains per checkpoint. “Across share” is the fraction of the tokens’ total squared norm that varies across rows: 1 is every row distinct, 0 is every row identical.

model	gene tokens, across share	strain summary, across share	prediction sd over strains	over compounds
transformer at initialization	0.073	0.005	0.011	0.001
transformer, prior 0, fold 0	0.0000	0.0000	0.0000	0.539
transformer, prior 0, fold 1	0.0002	0.0000	0.0000	0.119
transformer, prior 1, fold 0 (epoch 10 of 12)	0.399	0.562	0.031	0.418
embedding table, 5 folds	none	0.28	0.07 to 0.57	0.19 to 0.71

Without the prior, training drives all 6,607 gene tokens onto one vector: the strain summary is identical for every strain and the prediction is one value per compound. That is the round-1 symptom: train MSE flat at 0.60 (fold 0) and 0.975 (fold 1), within-compound raw Spearman near zero. Fold 0 of that arm was selected at 0.063 centered. The arm was cancelled after fold 1 (slurm 3017), and the genes-only floor with it (slurm 3019), since neither could be read while the encoder collapses.

With the prior at 1 the tokens stay distinct, and fold 0 was still improving at epoch 10 of 12 (test centered median 0.129, raw 0.024). Its prediction still varies 14 times more over compounds than over strains.

Hypothesis (untested): the encoder starts near collapse, with gene tokens at a mean cosine of 0.93 at initialization, and nothing but the graph prior pushes them apart, because a readout that can fit the compound effect alone gets most of its loss reduction without them.

Next arm, launched. The prior at 1 plus an identity skip: the strain’s genes summed from the encoder’s input embedding table, passed to the readout beside the transformer summary (`model.identity_skip`, slurm 3025). It asks whether the transformer adds anything once strain identity cannot be lost.

2026.09.29 - Model search: best compound-inductive model on Vanacloig alone

Question. Which model predicts the Vanacloig gene profile of a compound it has never seen, from the compound’s embedding alone, and does the cell graph transformer beat the simpler ones? Budget: 48 hours, two GPUs, beside the two round-1 GPU jobs (slurm 3016, 3030) that finish tonight.

Data. 143,218 cells: 3,598 queried genes by 41 compounds, 97% filled, one measurement per cell (mean of three biological replicates). Median served SE 0.138 against a response sd of 0.699. Median per-compound ceiling 0.834 (square root of the reliability index).

Protocol, fixed before any result.

- Compound-cold folds of `vanacloig_data.make_folds`, 5 folds, fold seeds 0, 1 and 2.
- Every choice is nested. The ladder picks options by leave-one-compound-out over the fold’s non-test compounds. The neural models pick their step on the fold’s 4 validation compounds. The test compounds never influence a choice.
- The primary target is centered Spearman, per compound, and the headline statistic is its median over the 41 compounds.

- Every model is compared PAIRED against two references on the same compounds, with a bootstrap interval over compounds (`compare_models.py`). The references are nested ridge on FCFP4 (`ladder:krr|linear:fcfp4_count`) and the ladder’s whole-pipeline pick (`ladder:selected`).
- Test centering for the ladder and the factorized models uses the mean over the fold’s non-test compounds. Round 1 (`train_vanacloig_cgt.py`, `baselines_same_folds.py`) centered on the 28 training compounds, so those numbers are not paired with round 2.

Rounds.

1. `baseline_ladder.py`: kernel ridge and nearest-neighbor maps over 12 compound embeddings and their kernels (slurm 3034).
2. `train_factorized.py`, `conf/factorized/r2_table.yaml`: a learned gene table times a compound encoder, bilinear or MLP head, 5 folds by 3 seeds plus the seed ensemble (slurm 3035).
3. The same trainer with the cell graph transformer as the gene encoder, `conf/factorized/r2_cgt.yaml`. One encoder pass per strain per step, not per cell.
4. Hypothesis to test if Vanacloig alone plateaus: pretrain the compound encoder on the other chemogenomic stores, excluding every Vanacloig compound. Wildenhain has 5,170 compounds on 242 genes, Hillenmeyer heterozygous has 303 compounds and Hoepfner 148 with InChIKeys. Wildenhain shares 10 of the 41 Vanacloig compounds, and Hoepfner and Hillenmeyer 2 each.

2026.09.29 - Audit of the ingestion, and what it changes for the scoring

A full reread of the paper, Piotrowski 2017 and the GEO raw counts against the loader and the cell table (issue #501 holds the report; issue #500 the strain background). The parquet reproduces the loader’s formula exactly, the sign is right (benomyl TUB3 is the most depleted strain), `response_ses` is the replicate SD over $\sqrt{3}$, and every InChIKey matches its name. Three findings bear on modeling:

- Nine served compounds were never reported by the paper, and their replicate reliability is near or below zero (sodium glyoxylate -0.91, sodium butyrate -0.43). They stay in the folds; `compare_models.py` reports every model over all 41 and over the 32 published compounds (`vanacloig_data.UNREPORTED_COMPOUNDS`).
- The loader normalizes by library size (CPM) where the paper used TMM. That leaves a compound-wide offset (crystal violet -2.8 log2 units) that a within-compound Spearman does not see; per-compound rank agreement with an edgeR reconstruction is 0.975 median.
- DMSO-delivered compounds carry the vehicle’s own profile (r 0.58 to 0.67 with the DMSO column) and no DMSO in their environment. Which compounds those are is in the unmirrored Table S1.

2026.09.29 - Round 2 results: nested ridge on FCFP4 is the bar, and nothing simple beats it

`compare_models.py` over `results/ladder/ladder_r2_scores.csv` (slurm 3034) and `results/factorized/r2_table` (slurm 3035). Centered Spearman, median over 123 compound-evaluations (41 compounds by fold seeds 0, 1, 2) for the ladder and 41 (fold seed 0) for the table models. “vs ridge” is the paired mean difference on the same compounds with a bootstrap 95% interval.

model	median, all 41	median, 32 published	vs ridge (95% CI)
ridge on FCFP4 counts, nested penalty	0.303	0.343	reference
kernel ridge, RBF on FCFP4	0.297	0.327	+0.006 (-0.014, +0.027)
kNN over the mean of all 12 RBF kernels	0.298	0.326	+0.002 (-0.024, +0.031)
ladder’s own nested pick, per fold	0.269		-0.018 (-0.044, +0.009)
best pretrained embedding (kNN, ChemBERTa-2 MTR)	0.291	0.312	-0.022 (-0.059, +0.015)
gene table x compound MLP, raw FCFP4, seed ensemble	0.287	0.301	-0.003 (-0.052, +0.046)
the same on a 64-dim PCA of FCFP4	0.233		

Every other embedding, kernel and table variant is below ridge, most with intervals clear of zero (`results/compare_models.csv`). The ladder's per-fold pick loses to fixed ridge because the inner leave-one-compound-out score over 32 compounds is itself noisy enough to pick a worse model about as often as a better one.

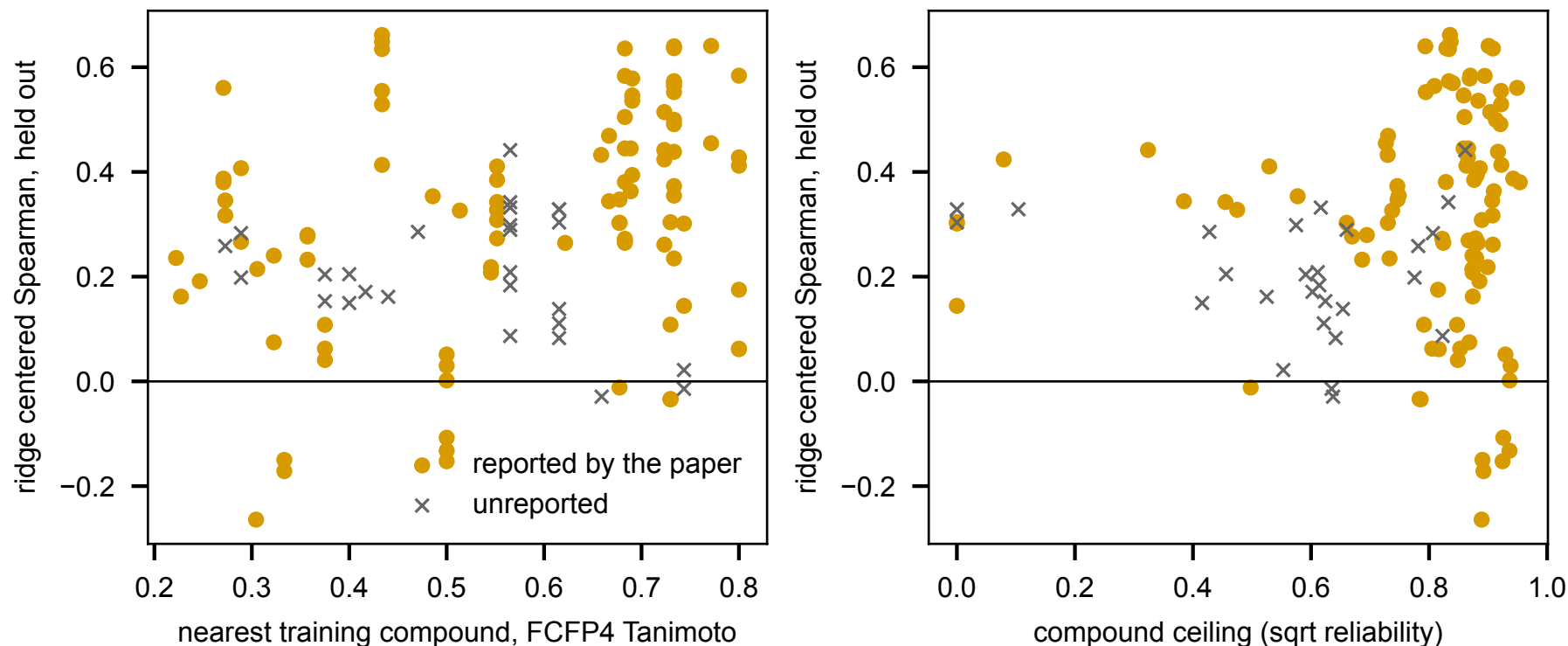
What this says: with 28 to 32 training compounds the compound side is the limit, and a 2,048-dimensional count fingerprint under ridge shrinkage is as good a compound representation as any pretrained embedding here. The gene side is not the limit: every gene is seen in training. So the cell graph transformer, which acts on the gene side, is not expected to move this number (hypothesis, being measured in round 2c, slurm 3038), and the lever is more compounds (round 3, slurm 3040).

2026.09.29 - Rounds 3 to 5, the transformer verdict, and the input audits of the other stores

Transformer, five folds (`r2_cgt`, slurm 3038). Cell graph transformer as the gene encoder of the factorized model, graph prior 1: centered median 0.061, paired -0.229 vs ridge (CI -0.29 to -0.17, 5 wins of 41). With the identity skip 0.030. The round-1 arm with the transformer unchanged (slurm 3016) scored 0.134, 0.002 and 0.043 on folds 0 to 2. The identity-skip round-1 job (3030) was cancelled after fold 0 (0.040, partial).

Other stores as auxiliary training (`r3`, `r3b`, slurm 3040, 3042). Shared compound encoder and gene table, per-source bias and adapter, every Vanacloig compound removed from the other stores. Over 123 compound-evaluations Wildenhain (5,160 compounds) is -0.008 vs ridge (CI -0.042, +0.026); its plain mean with ridge +0.013 (CI -0.013, +0.038). Hillenmeyer and Hoepfner as sources are below that. Graph smoothness on the shared gene table (`r4`, slurm 3046) is -0.02 to -0.05. A predicted profile in another store as a compound embedding (`phenotypic_embeddings.py`) is +0.003 at best; inside Wildenhain, structure predicts the profile's first principal direction at r 0.39 out of fold, the median direction at 0.11.

Coverage, not model (`similarity_vs_score.py`, slurm 3043). The ridge score of a held-out compound tracks its nearest training compound's FCFP4 Tanimoto (Spearman +0.29, n 123, p 0.001). Gamma-valerolactone (-0.20), methylglyoxal (-0.07) and MMS (-0.03) have ceilings above 0.89 and no analog.



Input audits of the other stores (report-only agents, issues carry the `dataset` label): Wildenhain #504 (33 of 242 strains are essential genes served as haploid deletions; the served release is the uncited 2016 Scientific Data extension; z is relative to the strain’s own median well), Hillenmeyer #505 (no strain background; HIS3, LYS2, MET15 dosage wrong on the BY marker loci; 33 arrays where the matrix header and the key file disagree; minimal medium served as plain SD; merged strains), Hoepfner pending. Vanacloig: #500 and #501.

Round 6, launched (`train_hit.py`). Where the molecule enters: readout only, the compound attending over all gene nodes (a hit distribution), the hit distribution propagated over the nine networks (the ego net around the molecule’s targets), and the deleted gene attending over the compound as one more perturbation token; each with 0 or 2 layers of message passing (`r6_mix`, slurm 3053), and a sizing sweep over width, depth, heads, hops and temperature (`r6_size`, slurm 3054).

2026.09.30 - Round 6: where the molecule enters, over 123 compound-evaluations

`train_hit.py`, fold seeds 0 to 2 (slurm 3053, 3057, 3058), sizing on fold seed 0 (3054). Centered Spearman; “vs ridge” is the paired mean difference on the same compounds with a bootstrap 95% interval; each arm is the 3-seed ensemble.

how the molecule enters	message passing	median	vs ridge
attends over all genes, hit mass propagated	2 layers	0.278	-0.011 (-0.037, +0.015)
2 hops on the nine networks (<code>hit_prop</code>)			
deleted gene attends over {itself, compound, null sink} (<code>gene_attend</code>)	2 layers	0.282	-0.014 (-0.044, +0.017)
readout only	2 layers	0.267	-0.019 (-0.045, +0.008)

how the molecule enters	message passing	median	vs ridge
attends over all genes, no propagation (hit)	2 layers	0.264	-0.025 (-0.054, +0.003)
hit_prop	none	0.248	-0.023 (-0.049, +0.002)
hit	none	0.265	-0.029 (-0.054, -0.004)
readout only	none	0.252	-0.030 (-0.055, -0.004)
gene_attend	none	0.251	-0.031 (-0.062, +0.002)

Two layers of message passing over the union of the nine networks lift every mixing by 0.01 to 0.02, from significantly below ridge to indistinguishable from it. Among the mixings at two layers the differences are within 0.01 and inside every interval; the molecule hitting genes and propagating is the best point estimate, and it is the only arm that matched ridge on a whole fold seed (0.304 on fold seed 1). Sizing on **hit_prop**: dim 64, two layers, eight heads, two hops is the best cell; one head (0.214), one hop (0.230), one layer (0.256), dim 128 (0.237), temperature 0.3 (0.222) and hidden 512 (0.238) are all below it (fold seed 0, ensembles).

Stacks (plain mean of saved predictions, `stack_predictions.py`), 123 evaluations: ridge plus the ten-seed gene table +0.013 (-0.004, +0.030; 70 of 123 wins); ridge plus **gene_attend** +0.012 (-0.011, +0.035); ridge plus **hit_prop** +0.005; ridge plus Wildenhain +0.013 (-0.013, +0.039). The ten-seed gene table alone matches ridge (+0.001).

Round 7, launched: ten-seed ensembles of the three level-2 mixings on every fold seed (**r7a**, slurm 3064) and propagation over one network family at a time: physical, coexpression, regulatory plus TFLink, STRING experimental plus database (**r7b**, slurm 3065). The launcher now packs four configs per GPU, since each model uses 2 GB and left the card idle.

2026.09.30 - Round 7: ten-seed ensembles and which network carries the hit; the search converges

Slurm 3064 (**r7a**, ten seeds per fold, every fold seed) and 3065 (**r7b**), 123 compound-evaluations each, paired vs ridge.

model	median	vs ridge
gene attends over compound, 2 layers, 10 seeds	0.297	-0.006 (-0.036, +0.025)
readout, 2 layers, 10 seeds	0.271	-0.009 (-0.036, +0.017)
hit and propagate, 2 layers, 10 seeds	0.277	-0.010 (-0.038, +0.014)
hit mass propagated over regulatory + TFLink only	0.276	-0.007 (-0.033, +0.017)
over coexpression only	0.281	-0.013 (-0.040, +0.014)
over physical only	0.276	-0.014 (-0.039, +0.011)
over STRING experimental + database only	0.264	-0.020 (-0.049, +0.007)
ridge + gene-attention ensemble, plain mean	0.315	+0.013 (-0.010, +0.037)
ridge + ten-seed gene table, plain mean	0.313	+0.013 (-0.004, +0.030)
ridge + three neural ensembles, plain mean	0.315	+0.009 (-0.013, +0.033)

Ten seeds instead of three move each arm by under 0.01. Which network carries the molecule’s hit mass makes no measurable difference. The best anything reaches is a plain mean of ridge with a neural ensemble, +0.013 with an interval that includes zero.

Where this leaves the question. On Vanacloig alone, compound-cold, every model family tried (twelve compound embeddings under kernel ridge and nearest neighbors; gene tables; the cell graph transformer; four ways of letting the molecule reach the genes, with and without message passing over the nine networks; auxiliary training on four other stores; phenotypic embeddings; graph smoothness; seed ensembles and stacks) lands within 0.03 of nested ridge on the raw FCFP4 counts, and none clears it. The two measured facts that explain it: with 28 to 32 training compounds the score of a held-out compound tracks its nearest training analog, and even inside Wildenhain’s 5,160 compounds structure predicts a chemical-genetic profile weakly. The gains available are in the data: more compounds in the same assay, a correctly defined input (#500, #501, #504 to #507), and a dose axis.

2026.09.30 - Round 8 setup: the transformer verdict was a training defect, not a verdict

The user’s reading of diagram 1 in [[experiments.035-env-chemgen-vanacloig-cgt.mermaid.cgt-compound]]: a molecule added to the medium is a perturbation of the cell state, a Type I operator like the deletion operator, and the factorized model never lets it be one; the compound meets the strain only at the bilinear head. Before building that operator (stage 2), the 0.061 of round 2 had to be understood, and two measurements say it was never a converged number.

Not converged. Round 2’s history (results/factorized/r2_cgt/cgt_bil_prior1_history.csv, replayed to W&B by wandb_replay_history.py): validation centered Spearman 0.004 at epoch 1, 0.044 at 10, 0.097 at 20, monotone to the last checkpoint; the selected step was 1,026 to 1,140 of 1,140 in four of five folds; train loss 0.72 to 0.44 and still falling. The flattening of the last three evals coincides with OneCycle annealing to zero. The table model gets 3,000 full-batch epochs; the transformer got 20.

The graph prior dominated the loss. Smoke smoke_r8.yaml (slurm 3071, 8 layers, 2 epochs) with the new per-step logging: penalty 23,256 and 46,815 against a data loss of 0.6 to 0.8; gradient norm 6,922 to 8,529 under a clip of 10, so the task gradient reached the weights scaled by about 1/700. default.yaml set graph_reg_lambda: 1.0 after the 025 sweep, where the data loss was over millions of pairs; 010 trains at 0.001. In round 2’s own history the penalty fell 39,734 to 613 while validation crept up: the model learned the graph prior first and the task as the prior allowed. The one-layer arm, in addition, had no prior at all because the regularized layer is index 1 (fixed in build_encoder: layer min(1, L - 1), per-head lambda 1, the single scale cgt_lambda).

Footprint (slurm 3071). 8 layers at 128 strains per step: 31.4 GB allocated, 34 s per epoch; at 64 strains 27.7 GB, 53 s. 1 layer: 8.4 GB, 10.5 s. A card is 44.4 GiB and a process holds 4 to 5 GB over its allocation (jobs 3072 and 3073 OOM’d packing 8 + 1 layers), so the 8-layer arms run alone and the small arms two per card.

Round 8 (r8_small_{a,b}.yaml, slurm 3074 and 3075; r8_deep_{a,b}.yaml, 3078 and 3079 chained behind them): layers in {1, 2, 8} at 9 heads, prior weight in {0, 0.001, 1}, 200 epochs, one fold per process, fold seed 0, every epoch logging train loss, penalty, gradient norm, learning rate, validation and held-out centered Spearman, the per-compound validation scores, wall time and peak memory. First live curves (partial, in flight): L1_lam0_f0 held-out 0.292 at epoch 9 and 0.258 at 73 with train loss 0.061; L1_lam1e-3_f0 0.288 at epoch 30; the cancelled L8_lam0_f0 0.012 at epoch 9. Depth slows learning on its own, so the 8-layer chain was trimmed to two folds per weight.

Stage 2, the environment operator (head: operator, EnvironmentOperator in train_factorized.py; CPU smoke smoke_env.yaml passed): the compound token e_c from the fingerprint MLP acts on every gene of the strain’s post-deletion state $H \in \mathbb{R}^{B \times N \times d}$ through a per-gene, per-head sigmoid gate, $H + \beta a \odot v_c$, with β a ReZero scalar at zero so the model starts as the identity; the readout is invariant over the genome, the deleted rows summed, the mean over genes and the token, through an MLP beside the gene bias and compound offset. Both readouts are linear in the update, so the pooled term is $\bar{a} v_c$ and the $B \times C \times N \times d$ state is never materialized. Round 9 (r9_operator.yaml, r9_control.yaml, slurm 3080 and 3081 chained behind the deep chains): operator versus the bilinear head on the same one-layer, prior-0.001 encoder, 100 epochs, five folds on each of fold seeds 0 to 2, for the 123-evaluation paired comparison against ridge.

2026.10.02 - Rounds 8 to 13: trained to saturation, the cell graph transformer ties ridge

Metric throughout: for each held-out compound, the Spearman correlation between predicted and measured response across the ~3,500 strains after subtracting each strain’s mean over the fitted compounds from both sides (centered Spearman). “Compound-evaluations” counts held-out compounds over splits; all three fold seeds give 123. Paired differences are against nested ridge on the same compound-evaluations, with a bootstrap 95% interval (compare_models.py, results/compare_models.csv). Curves, losses and wall times are from summarize_cgt_rounds.py (results/cgt_rounds_curves.csv, results/cgt_rounds_loss_curves.csv, results/cgt_rounds_loss_baselines.csv).

Correction to the section above. The prior at weight 1 was not what held round 2 back. With the prior off, eight layers do not train at all (train loss 0.62 at epoch 200, held-out 0.004); with the prior at weight 1 and 200 epochs the same model reaches the level of the shallow encoders. Round 2’s 0.061 was its 20-epoch budget. The prior is unnecessary only for encoders of six layers or fewer.

Depth and the graph prior (round 8). Fold seed 0, one seed, fit on the training compounds, epoch picked on the 4 validation compounds, 200 epochs, 9 heads, width 180. Slurm 3074, 3075, 3078, 3113, 3116, 3133, 3135 on two RTX 6000 Ada cards (44 GiB), two processes per card for the no-prior arms and one for the 8-layer arms.

layers	prior weight	folds (compound-evaluations)	held-out centered Spearman at the last epoch, mean over folds	mean paired difference in centered Spearman vs ridge, validation-selected epoch (95% CI)	train loss of the last batch at epoch 200 (standardized MSE)	minutes per fold	peak GPU memory (GB)
1	0	5 (41)	0.259	-0.048 (-0.104, +0.012)	0.028	45	7.7
2	0	5 (41)	0.254	-0.072 (-0.130, -0.015)	0.032	44	7.7
3	0	2 (17)	0.256	+0.011 (-0.038, +0.060)	0.035	52	7.8
4	0	5 (41)	0.268	-0.092 (-0.159, -0.023)	0.037	50	7.9
6	0	2 (17)	0.262	-0.011 (-0.077, +0.051)	0.050	49	8.0
8	0	2 (17)	0.004	-0.272 (-0.372, -0.172)	0.622	38	8.1
1	0.001	5 (41)	0.270	-0.053 (-0.112, +0.005)	0.057	108	13.2
2	0.001	5 (41)	0.263	-0.086 (-0.139, -0.028)	0.042	122	15.8
4	1	5 (41)	0.271	-0.048 (-0.099, +0.006)	0.091	100	20.9
8	1	5 (41)	0.262	-0.060 (-0.111, -0.008)	0.088	124	31.5

The last-epoch column is the one that isolates depth: 0.25 to 0.27 for every arm that trains. The validation-selected column ranges from +0.011 to -0.092 over the same arms because each fold picks its epoch on 4 compounds; four layers without the prior has one of the best last-epoch numbers and the worst selected one. Shallow encoders plateau by epoch 20 to 40; the 8-layer encoder with the prior climbs to about epoch 160. Depth costs time and memory and buys nothing on this dataset, and the prior more than doubles the cost per fold.

Protocol: pool fit, fixed epochs, seeds (round 9 control). The selector, not the model, was costing the score. The round-9 control is the one-layer, no-prior encoder with the bilinear head, fit on the whole non-test pool (as ridge is), a fixed 50 epochs keeping the last, three seeds averaged, on all three fold seeds (slurm 3114). It moves the same encoder from -0.048 (41 compound-evaluations) to -0.007 (-0.028, +0.015) on 123.

How the compound enters (rounds 9 and 10). Three heads on that encoder under that protocol; diagrams of the first in [[experiments.035-env-chemgen-vanacloig-cgt.mermaid.cgt-compound]]. The bilinear head multiplies the strain vector with the compound vector at the readout. The operator (`EnvironmentOperator`) gates the compound’s value onto every gene of the strain’s post-deletion state, each gene independently. The environment encoder (`EnvironmentEncoder`) puts the compound token in front of the 6,607 wildtype gene tokens for one further transformer layer, so genes take from the compound and from each other after it, and reads the strain at its deleted genes’ rows. Slurm 3080, 3114, 3134.

compound enters as	compound-evaluations	median centered Spearman per held-out compound	mean centered Spearman per held-out compound	mean paired difference in centered Spearman vs ridge (95% CI)	compound-evaluations won vs ridge	mean paired difference vs the bilinear head (95% CI)
bilinear factor at the head	123	0.285	0.285	-0.007 (-0.028, +0.015)	55	reference
gate on each gene (operator)	123	0.270	0.266	-0.025 (-0.066, +0.015)	53	-0.019 (-0.053, +0.014)
token the genes attend to (environment encoder)	123	0.281	0.300	+0.008 (-0.026, +0.045)	56	+0.015 (-0.014, +0.047)

Treating the molecule as a perturbation of the cell state does not separate from treating it as a factor at the head: the gate is slightly behind, the token slightly ahead, both intervals through zero. The token version is the only single neural model with a positive paired mean against ridge.

Stacks. Plain mean of ridge and a model’s saved prediction, no fitted weight (`stack_predictions.py`).

stack	compound-evaluations	median centered Spearman per held-out compound	mean paired difference in centered Spearman vs ridge (95% CI)	compound-evaluations won vs ridge
ridge + environment encoder	123	0.334	+0.028 (-0.001, +0.059)	66
ridge + operator	123	0.321	+0.013 (-0.014, +0.043)	59
ridge + ten-seed gene table (round 5)	123	0.313	+0.013 (-0.003, +0.032)	70
ridge + bilinear head	123	0.300	+0.002 (-0.017, +0.020)	62

The ridge and environment-encoder stack is the best result of the experiment and the closest any model has come to clearing ridge. The same stack with the bilinear head gains nothing, so the environment encoder’s errors are less like ridge’s.

Loss curves and what dominates the loss (rounds 12 and 13). Until round 11 the trainer logged rank scores and the loss of one batch. It now logs, every epoch, the standardized MSE over every strain on the fitted, validation and held-out compounds. Round 12 (slurm 3165) is the one-layer, no-prior arm fit on the training compounds only, five folds, so its validation loss is out of sample.

epoch	train loss, standardized MSE over the fitted compounds (mean over 5 folds)	validation loss, standardized MSE on 4 never-fitted compounds	held-out loss, standardized MSE	validation centered Spearman	held-out centered Spearman
1	0.763	2.448	1.669	-0.001	0.004
10	0.287	2.436	1.515	0.224	0.258
20	0.127	2.426	1.513	0.233	0.257
40	0.061	2.410	1.523	0.241	0.264
60	0.037	2.364	1.539	0.242	0.258
100	0.022	2.401	1.537	0.243	0.256

The model memorizes the fitted compounds (0.76 to 0.02) while the validation loss stays at 2.4 and the held-out loss moves from 1.67 to 1.54. Rank order on new compounds is learned within ten epochs and then holds; magnitudes are not learned. Held-out loss by fold against constant predictors, from the saved predictions:

fold (fold seed 0)	held-out loss (standardized MSE), global mean	held-out loss, each gene's mean	held-out loss, nested ridge	held-out loss, 1 layer val-selected	held-out loss, 8 layers prior 1 val-selected	held-out loss, bilinear pool fit 3 seeds	held-out loss, environment encoder pool fit 3 seeds
0	0.386	0.430	0.386	0.365	0.372	0.404	0.384
1	5.837	5.760	5.400	5.330	5.473	5.343	4.979
2	0.402	0.358	0.284	0.374	0.371	0.337	0.300
3	0.543	0.500	0.476	0.962	1.118	0.484	0.456
4	0.577	0.534	0.414	0.467	0.418	0.423	0.497
mean	1.549	1.516	1.392	1.500	1.551	1.398	1.323

The best model lowers held-out MSE by 13% against predicting each gene's mean. One compound sets the level: crystal violet has 14.1 times the median compound's response variance and 24.7% of the total (`results/compound_variance_share.csv`); it is in the test set of fold 1 and in the validation set of fold 3, which is where the two validation-selected arms lose (0.96 and 1.12 against ridge's 0.48). It is also the compound with the normalization offset in issue #501.

Round 13 (slurm 3168) tested the consequence: z-score each fitted compound's profile across strains before the fit (`target_scale: per_compound`), otherwise the round-9 control. Null: median centered Spearman 0.294 against the control's 0.285, mean paired difference against the control -0.004 (-0.020, +0.010), 63 of 123 won; against ridge -0.011 (-0.036, +0.015). Crystal violet's own held-out centered Spearman is lower with scaling on every fold seed (control 0.546, 0.523, 0.556; scaled 0.471, 0.462, 0.428). The dominant compound in the loss was not costing rank accuracy on the others.

W&B. Saved view, runs grouped by arm, 100 runs per panel, built by `wandb_view.py`:

https://wandb.ai/zhao-group/torchcell_035-env-chemgen-vanacloig-cgt?nw=sy9905pud6q

The validation panels plot only arms whose validation compounds were never fitted; an arm fit on the pool scores about 0.7 on them because it trained on them. Group pages follow `/groups/<arm>` with the arm names of `results/wandb_views.json`.

Where this leaves the transformer question. Trained to saturation, the cell graph transformer as the gene encoder ties nested ridge on Vanacloig alone at any depth from one to eight layers, and no way of injecting the compound separates from another. The best number is a plain mean of ridge with the environment encoder, +0.028 with an interval that reaches -0.001. Round 11 (nine seeds of the environment encoder and of the bilinear control, slurm 3166 and 3169) is the remaining measurement on whether that gap is real.

2026.10.02 - Rounds 11, 14 and 15: nine seeds, a second environment layer, and the encoder's loss curves

Scores are the centered Spearman per held-out compound across about 3,500 strains, 123 compound-evaluations (41 compounds on each of fold seeds 0, 1, 2), paired against nested ridge on the same evaluations. All numbers are from `results/compare_models.csv` and `results/compare_pairs.csv` (`compare_models.py`).

Two intervals, and which one to quote. `compare_models.py` now reports two 95% bootstrap intervals for a paired mean difference. The first resamples the 123 compound-evaluations, which treats the three fold-seed evaluations of one compound as independent; they are not, so it is too narrow on pooled rows. The second resamples the 41 compounds, each carrying its mean difference over fold seeds. The compound interval is the one to quote, and it applies to every pooled interval written in the sections above: none of them excluded zero, so no conclusion changes, but each is wider than written.

Round 11: nine seeds each (slurm 3166, 3169). Six more seeds of the environment encoder and of the bilinear control, so each arm has nine (three blocks of three). A stack member written as several blocks is averaged within itself first, then averaged with ridge (`stack_predictions.py`).

model (every fold seed pooled)	median centered Spearman (123 compound-evaluations)	mean centered Spearman (123)	mean paired difference vs nested ridge	95% CI resampling evaluations	95% CI resampling the 41 compounds	evaluations won vs ridge (of 123)	compounds improved vs ridge (of 41)
ridge + environment encoder, nine seeds	0.334	0.323	+0.032	(+0.003, +0.062)	(-0.007, +0.078)	71	22
ridge + environment encoder + bilinear, nine seeds each	0.326	0.321	+0.030	(+0.005, +0.056)	(-0.001, +0.066)	70	25
environment encoder alone, nine seeds	0.289	0.301	+0.010	(-0.027, +0.047)	(-0.042, +0.069)	63	18
ridge + bilinear, nine seeds	0.289	0.295	+0.004	(-0.014, +0.022)	(-0.020, +0.027)	65	22
bilinear alone, nine seeds	0.282	0.288	-0.003	(-0.024, +0.019)	(-0.031, +0.025)	59	21
nested ridge	0.303	0.293	0				

Seeds saturate at three: the environment encoder alone is +0.008 with three seeds, +0.009 with six and +0.010 with nine; its stack with ridge is +0.028, +0.031 and +0.032. On the 32 published compounds the best stack is +0.022 with a compound interval of (-0.023, +0.081). No model or stack has a compound interval that excludes zero.

Round 14: a second environment layer (slurm 3211, 3212). Two transformer layers over the compound token and the gene tokens instead of one, otherwise round 10 (`env_layers: 2`, three seeds, 50 epochs, pool fit). Null: median centered Spearman 0.318 against 0.281 for one layer, mean 0.292 against 0.300; mean paired difference two layers minus one -0.008, compound interval (-0.034, +0.015), 63 of 123 evaluations won, 21 of 41 compounds improved. Against ridge the two-layer model is +0.000 (-0.051, +0.058). A read at 33 evaluations had it at +0.020; that was the first four folds.

Round 15: the encoder’s out-of-sample loss curves (slurm 3221). The environment encoder fitted on the training compounds only (one seed, fold seed 0, five folds), so its validation loss is out of sample; the counterpart of round 12 for the bilinear head. Mean over the five folds, from `results/cgt_rounds_loss_curves.csv` (`summarize_cgt_rounds.py`):

epoch	bilinear: train loss (standardized MSE)	bilinear: validation loss	bilinear: held-out centered Spearman	environment encoder: train loss (standardized MSE)	environment encoder: validation loss	environment encoder: held-out centered Spearman
1	0.763	2.448	0.004	0.760	2.452	0.000
5	0.740	2.601	0.120	0.745	2.703	0.001
10	0.287	2.436	0.258	0.597	2.467	0.166
20	0.127	2.426	0.257	0.386	2.350	0.189
30	0.084	2.416	0.259	0.279	2.333	0.243
40	0.061	2.410	0.264	0.237	2.301	0.245
50	0.047	2.410	0.259	0.229	2.284	0.247

The two heads differ in kind. The bilinear head fits the training compounds almost exactly (0.047) and its validation loss is flat from epoch 10. The environment encoder ends its 50 epochs at a train loss of 0.229 with the validation loss still falling, under a one-cycle schedule that has taken the learning rate to zero by then. Fifty epochs is therefore not saturation for the environment encoder; round 16 trains it for 150.

2026.10.02 - Rounds 16 and 17 setup: the encoder at 150 epochs, and the learning curve in fitted compounds

Two questions follow from the rounds above, one per card (GilaHyper, RTX 6000 Ada, PARALLEL=2).

Round 16 (slurm 3231, `conf/factorized/r16_envenc_e150.yaml`): the environment encoder of round 10 for 150 epochs instead of 50, one seed, five folds on each of fold seeds 0, 1, 2. The paired reference is the seed-0 member of round 10 on the same folds, which differs in the epoch budget only (the one-cycle warmup stretches with it, 7.5 epochs instead of 2.5). It is the pair `r16_envenc_e150:enc_L1_lam0_e150:seed0` against `r10_envenc:enc_L1_lam0:seed0` in `results/compare_pairs.csv`.

Round 17 (slurm 3232, `conf/factorized/r17_curve.yaml`): the learning curve. Every model so far was fitted on the 32 or 33 non-test compounds of a fold. Each fold's pool is cut to nested subsets of 8, 16 and 24 compounds (`vanacloig_data.subsample_pool`, config field `n_fit_compounds`), the environment encoder is fitted on each (one seed, 50 epochs), and the fold's held-out compounds are scored with each side centered by its mean over the fitted compounds. Nested ridge on the same subsets is `learning_curve.py --ridge` (slurm 3233, complete); its whole-pool fit reproduces the saved ladder prediction, which the script asserts.

Ridge alone (`results/learning_curve/ridge_curve.csv`), 123 compound-evaluations per row:

fitted compounds	ridge: median centered Spearman per held-out compound (123 compound-evaluations)	ridge: mean centered Spearman (123)	ridge: median raw Spearman (123)	ridge: mean raw Spearman (123)
8	0.202	0.196	0.153	0.173
16	0.257	0.240	0.226	0.217
24	0.299	0.278	0.246	0.249
32 or 33 (whole pool)	0.303	0.291	0.273	0.295

The mean centered score rises 0.044 from 8 to 16 compounds and 0.051 from 16 to the whole pool, about 0.05 per doubling with no sign of a plateau at 33 compounds. The centered target itself changes with the subset (the mean it subtracts is over fewer compounds), so the raw column, whose target is fixed, is the check; it rises the same way. Hypothesis (untested): the same slope continues past 41 compounds of this assay. Whether compounds from OTHER assays move this curve is a separate question; the gene-table multi-source arms of round 3 did not gain on ridge (-0.008 to -0.035, intervals through zero), under the pre-round-9 protocol.